A Provenance-First Tax Workpaper

Core Idea & Data Model

MiraFlores AI - AI Engineer (Tax) case study - Candidate: Shashi Kumar Sheelam. All household data is fictional, per the packet.

The problem with a spreadsheet clone

A computed tax return is a set of numbers whose only defense is where each number came from. A spreadsheet shows values; when a reviewer asks why a line is 212,400, the answer lives in someone's memory or on page 19 of a 32-page 1099. Review becomes archaeology - and errors survive because nobody can afford to re-derive every line.

Core idea: every return line is a node in an evidence graph

return line -> inputs (other return lines) -> sources (extracted values: field, value, confidence, highlight position) -> simulated source document (page text as delivered).

Clicking any line walks that graph to the simulated source document with the exact field highlighted, drawn from the packet's stored position box. A number is never more than one click from its evidence. Simulated documents sit behind a clean document-view interface, so real page images can replace them without touching the rest of the system.

- Checks are independent of the engine. The workpaper recomputes what it can check without trusting the preparer's engine: extraction-confidence triage, completeness (every document used or explicitly dispositioned), leaf lines with no source, organizer-vs-document identity matching, prior-year variance. Engine diagnostics and independent findings are labelled separately.
- Adjustments never overwrite. A correction is a new fact - original value, proposed value, author, reason, tax-effect category, and an estimated tax effect priced from the version's tax_effect_rates. A sign-off records the line's value at signing time; if the value changes, its effective status recomputes to reopened on read.
- The packet is immutable; workpaper state is separate. Notes (which can block filing), sign-offs, adjustments, verifications and a sensitive-data view log live in a separate store keyed by version and line. Deleting it resets the workpaper; the evidence never changes.

Data model (packet, read-only)

Document (74): id, type, issuer, pages[]; extraction.status = 68 extracted / 2 partial / 4 not extracted. Extracted value (3,012): id, document_id, page, field, value, confidence, position {x0,y0,x1,y1} - the atomic unit of evidence. Return version (v1-v3): computed_at, trigger, summary, tax_effect_rates, diagnostics, changes_from_previous; v3 (2026-06-23) is current, 627 lines. Return line: key, form/line, value, rule, inputs[], sources[] - the graph node. Client layer: 45 questionnaire answers, messages, call notes (claims to check, not facts). Firm layer: 6 staff with roles and sensitive_data_clearance, 29 timeline events, 5 what-if alternatives.

Permissions, what-ifs, timeline - honesty by design

Permissions are enforced on the server, not the switcher: every request carries the acting person's id and the server resolves the real role from the packet's staff list. Staff with write roles may sign off, adjust and verify; the client may read and ask questions but cannot sign off or see firm-internal notes. SSNs and bank numbers unmask only for clearance holders (a preparer with a pending background check sees them masked), and every sensitive view is logged. What-ifs: the engine cannot be rerun, so exploration uses only the 5 precomputed alternatives and tax-effect rates, with combined views labelled as sums, not engine results. Timeline: the 29 events replay in order; once signed (Form 8879) and e-file accepted, the filed state freezes and later events form an amend-on-top layer.

Disclosure: this prototype and document were built with AI assistance (Muse), as the case study explicitly permits when disclosed. Full disclosure text and deliverables map: SUBMISSION.md.